
GNN-Based BERT for Understanding Context from Music

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Abstract

We investigate the feasibility of using graph representations of musical structure in addition to pretrained text encoders to aid music understanding. From each audio clip, we build a segment graph whose nodes are chroma summaries of fixed-length segments and whose edges connect temporally adjacent segments and segments with similar harmonic content. We test three setups, with increasing difficulty: (i) a BERT classifier that completes MagnaTagATune tag sets based on partial tag observation, (ii) a GNN genre classifier using only structure compared to a Mel-spectrogram CNN, and (iii) a GNN-BERT model that attends over BERT token states. BERT tag completion achieves a 0.74 macro-F1 score on 50 tags. The structure-only GNN is very weak (0.09 macro-F1, 36.6% accuracy over 16 genres) and significantly underperforms the CNN baseline (0.28 macro-F1, 49.1%), which we believe is due to the small number of nodes (≈ 4) in the graph and the 12-dimensional chroma node features. Despite this, the performance of the cross-attention fusion is still better than that of the BERT-only, GNN-only, and early-concatenation variants in macro-F1 (0.27 vs. 0.17, 0.10, and 0.21), suggesting that the two views bring complementary information. We use a single training run evaluated on validation splits, and discuss some of the limitations of the presented system, such as the sparse free-text tags in FMA, and describe an extension to the audio-graph-caption retrieval problem that is left for future work.

1 Introduction

Music can be explained in two distinct ways. One is the signal itself. The signal carries the structure of the music that cannot be explained through text, such as repeating sections, harmonic progressions, and contrasts between parts of a piece. The other is the textual context. The textual context contains the lyrics, tags, and reviews. Pretrained language models such as BERT [Devlin et al., 2019] strongly encode the second kind of information, while audio models, most commonly convolutional networks on spectrograms [Choi et al., 2016, Pons et al., 2018, Won et al., 2020], learn the first implicitly.

Music can easily be represented in a graph. Temporal adjacency connects adjacent segments and similarity connects distant segments when a clip is divided into segments, where they sound similar (e.g. a repeated chorus). This is the same insight into self-similarity analysis of songs [Foote, 2000, McFee and Ellis, 2014]. This structure can then be directly learned by Graph Neural Networks (GNNs) [Kipf and Welling, 2017, Hamilton et al., 2017, Veličković et al., 2018].

This report poses a simple question: *can a GNN over audio-structure graphs contribute information to a BERT text encoder, and what’s the best way to do so?* We organise the work as a sequence of tasks:

- **Task 1 (text only).** A fine-tuned BERT model predicts the full set of MagnaTagATune [Law et al., 2009] tags for a clip from a randomly masked subset of its own tags. We refer to this as *tag completion*.
- **Task 2 (structure only).** A GraphSAGE model classifies FMA-medium [Defferrard et al., 2017] tracks into 16 genres using only segment graphs, and is compared against a CNN on log-mel spectrograms.
- **Task 3 (fusion).** The graph embedding acts as a query over BERT token states in a cross-attention layer. We compare this with BERT-only, GNN-only and early-concatenation variants.
- **Task 4 (retrieval, not executed).** A contrastive dual encoder for audio-graph–caption retrieval on MusicCaps [Agostinelli et al., 2023]. We describe the design in Appendix C but report no results.

Our main findings are that (a) the segment graphs as currently built are a severe information bottleneck, so the structure-only GNN falls well below of the CNN baseline, and (b) despite this, cross-attention fusion is the best of the four Task 3 variants by macro-F1. The fusion variant of Task 3 is the highest-scoring variant based on macro-F1. We present these as preliminary results from single runs on validation data and are explicit about where the evidence is weak (Section 6).

2 Related work

Music auto-tagging and genre classification. The common method for music tagging is a CNN on spectrograms [Choi et al., 2016], and it has been scaled up and systematically compared on benchmarks such as MagnaTagATune [Pons et al., 2018, Won et al., 2020]. FMA [Defferrard et al., 2017] offers audio, genre hierarchies, and user-supplied metadata for a big collection of Creative Commons tracks, and is popularly used for genre recognition. Instead, we adopt a small CNN as our audio baseline rather than try to match these systems.

Structure and graphs in music. Self-similarity matrices expose repetition and novelty in audio [Foote, 2000]. McFee and Ellis [2014] have developed a graph-based temporal and long-range recurrence linkage model, and use spectral clustering to segment songs. Our segment graphs are similar, but we learn from them using GNNs: GraphSAGE [Hamilton et al., 2017] that inductively aggregates neighbour features, and graph attention networks [Veličković et al., 2018].

Multimodal fusion and audio–text learning. Cross-attention between modalities, such as tokens attending to each other [Vaswani et al., 2017], is prevalent in vision-and-language (VL) models, like ViLBERT [Lu et al., 2019]. For audio and text, Dual Encoders are used, like CLIP models [Radford et al., 2021], CLAP [Elizalde et al., 2023], or MuLan [Huang et al., 2022], which map both modalities into a shared space. Our fusion model is a single-query variant of cross-attention that is lightweight, and our proposed Task 4 is based on the recipe of contrastive.

3 Method

Figure 1 summarises the full model. Tasks 1 and 2 use its text and graph branches in isolation.

3.1 Audio features and segment graphs

Audio is decoded to mono at 22,050 Hz with librosa [McFee et al., 2015] (via ffmpeg when available). We compute a 12-bin chroma representation with hop length 512 and z-normalise each feature over time within the track. The track is split into non-overlapping fixed windows of 7.5 s, discarding the remainder, and each window is mean-pooled to a vector $\mathbf{x}_i \in \mathbb{R}^{12}$.

The segment graph $G = (V, E)$ has one node per segment. Its edge set is the union of temporal edges $\{(i, i+1), (i+1, i)\}$ and similarity edges $\{(i, j) : i \neq j, \cos(\mathbf{x}_i, \mathbf{x}_j) > \tau\}$ with $\tau = 0.8$, so all edges

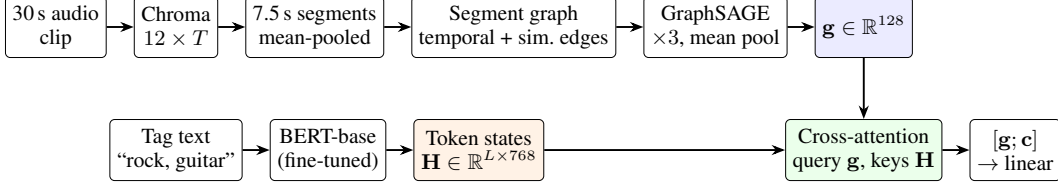


Figure 1: GNN-BERT fusion model (Task 3). An audio clip becomes a small segment graph encoded by GraphSAGE into $\mathbf{g}$; tag text is encoded by BERT into token states $\mathbf{H}$. The graph embedding attends over the tokens, and the concatenation of $\mathbf{g}$ with the attended text vector $\mathbf{c}$ is classified. Task 1 uses only the lower branch and Task 2 only the upper branch.

are undirected. Cosine similarity is stored as an edge attribute, although the GraphSAGE encoder below does not use it. For a 30s FMA clip this construction yields only four nodes (Section 6). The code also contains a chord-transition graph builder that was not used in the reported experiments.

3.2 Encoders

Graph encoder. We use three GraphSAGE layers [Hamilton et al., 2017] with mean aggregation and hidden size 128, with ReLU and dropout 0.2 between layers. Node embeddings are mean-pooled into a graph embedding $\mathbf{g} \in \mathbb{R}^{128}$. A GAT variant [Veličković et al., 2018] is implemented but was not used for the reported results.

Text encoder. We fine-tune bert-base-uncased [Devlin et al., 2019, Wolf et al., 2020] end to end with a maximum length of 256 tokens. It returns the [CLS] embedding $\mathbf{h}_{\text{CLS}}$ and the token states $\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_L]$, $\mathbf{h}_\ell \in \mathbb{R}^{768}$.

3.3 Cross-attention fusion

The graph embedding is projected to a single query that attends over the BERT tokens:

$$\mathbf{q} = \mathbf{W}_q \mathbf{g}, \quad \mathbf{k}_\ell = \mathbf{W}_k \mathbf{h}_\ell, \quad \alpha_\ell = \frac{\exp(\mathbf{q}^\top \mathbf{k}_\ell / \sqrt{d_a})}{\sum_{m \in \mathcal{M}} \exp(\mathbf{q}^\top \mathbf{k}_m / \sqrt{d_a})}, \quad \mathbf{c} = \sum_{\ell \in \mathcal{M}} \alpha_\ell \mathbf{h}_\ell, \quad (1)$$

where $d_a = 128$, $\mathcal{M}$ is the set of non-padding tokens, and the values are the unprojected token states. The fused representation $\mathbf{z} = [\mathbf{g}; \mathbf{c}] \in \mathbb{R}^{896}$ is mapped to class logits by a linear layer. The model therefore asks *which words in the text are most relevant to this piece’s structure*, and the attention weights can be inspected. The implementation includes an optional two-output head for valence and arousal regression; it was inactive in our runs because the Task 3 data provide no emotion targets.

Early concatenation baseline. As a fusion ablation we replace attention with $\mathbf{z} = [\mathbf{g}; \mathbf{h}_{\text{CLS}}]$ followed by the same linear head.

3.4 Audio baseline

For Task 2 we train a small CNN on log-mel spectrograms (128 mel bands, hop 512), padded or truncated to 1,300 frames (≈ 30 s). It has three 3×3 convolution blocks with 16, 32 and 64 channels, batch normalisation and ReLU, max-pooling after the first two blocks, global average pooling and a linear classifier.

4 Experimental setup

Task 1 data: MagnaTagATune. 50 most-used tags and clips with at least one tag are used. Fixed ID lists for train, validate, and test clips (18,706 / 1,825 / 5,329 before filtering). There are no captions in MagnaTagATune, and we make a pseudo-caption for each clip: we sample $\max(1, \text{round}(0.5 |T|))$ of its $|T|$ positive tags and concatenate them with a comma. The target is the complete 50 dimensional tag-vector. As the input comes from a subset of the target, Task 1 is designed to assess the ability of BERT to fill in missing tags using tag co-occurrence rather than free text.

Table 1: Validation results after 20 epochs (single run). Task 1 is 50-tag multi-label tag completion on MagnaTagATune; Tasks 2–3 are 16-class genre classification on FMA-medium, where micro-F1 equals accuracy. The structure-only GNN is far behind the CNN, but combining it with text through cross-attention gives the best macro-F1 among the Task 3 variants.

Task	Model	Macro-F1	Micro-F1 / Acc.	Mean AUC-PR
1	BERT tag completion	0.739	0.765	–
2	GNN (GraphSAGE, graph only)	0.086	0.366	–
2	CNN baseline (log-mel)	0.277	0.491	–
3	BERT only	0.171	0.405	0.249
3	GNN only	0.098	0.373	0.124
3	Early concatenation	0.206	0.447	0.291
3	Cross-attention (ours)	0.269	0.459	0.276

Tasks 2–3 data: FMA-medium. We have adopted the official FMA-medium training and validation splits as well as the FMA’s 16 top-level genres, thus resulting in a total of 2,505 validation tracks. The top-level genre is called the label. Task 3 text input consists of the free-form tags field in FMA’s `raw_tracks.csv` file, separated by commas, or “no tags” if it is empty. Tracks that cannot be decoded are replaced by the following decodable track in the split.

Training. All models are trained for 20 epochs using a batch size of 32 and AdamW [Loshchilov and Hutter, 2019] (weight decay 0.01). The learning rate for the pre-trained BERT parameters is 2×10^{-5} and 10^{-3} for the randomly initialised modules (GNN, CNN, fusion and heads). The linear head in Task 1 also takes 2×10^{-5} . It is binary cross-entropy with logits on each task, including upon application to one-hot labels for single-label genre tasks. The seed used to generate the random numbers is 42. It is implemented in PyTorch [Paszke et al., 2019], PyTorch Geometric [Fey and Lenssen, 2019], and Hugging Face Transformers [Wolf et al., 2020]. The complete hyperparameters are provided in Appendix A. Hyperparameters were chosen after brief manual trials on the validation set.

Evaluation. For Task 1, a tag is predicted if the sigmoid probability is greater than 0.5. The predicted genre for Tasks 2-3 is the argmax. We report macro-F1 (the unweighted mean of F1 per class) and micro-F1. In the single-label genre tasks, micro-F1 equals accuracy, so we label it as such. For task 3, mean per-class average precision (AUC-PR) based on probabilities is also reported. **All numbers are measured on the validation split after the final (20th) epoch of a single run;** no test-set results are reported (Section 6).

5 Results

5.1 Main results

Task 1: BERT completes tag sets well. BERT reaches 0.739 macro-F1 and 0.765 micro-F1 (Table 1), up from 0.49 macro-F1 after the first epoch, and the curve plateaus from about epoch 16 (Appendix B). Inspecting individual predictions shows the model mostly reproduces the visible tags and adds strongly co-occurring ones. For the input “electronic, rock” it predicts {techno, electronic, rock} against the target {electronic, rock, fast}. For “vocals, male” it predicts {male, vocals} but misses {guitar, vocal, male vocal}. Its misses in these examples include near-synonyms such as “vocal”/“vocals”, which are separate classes in MagnaTagATune. Because part of the answer is always visible in the input, these scores are an upper bound on text-only performance and should not be compared with published audio tagging results.

Task 2: structure-only graphs underperform an audio CNN. The GNN reaches only 0.086 macro-F1 and 36.6% accuracy. The CNN baseline reaches 0.277 and 49.1% after the final epoch. The CNN’s validation scores also fluctuate considerably between epochs: it peaks at epoch 16 with 0.361 macro-F1 and 65.7% accuracy, then falls. The GNN’s very low macro-F1 alongside moderate

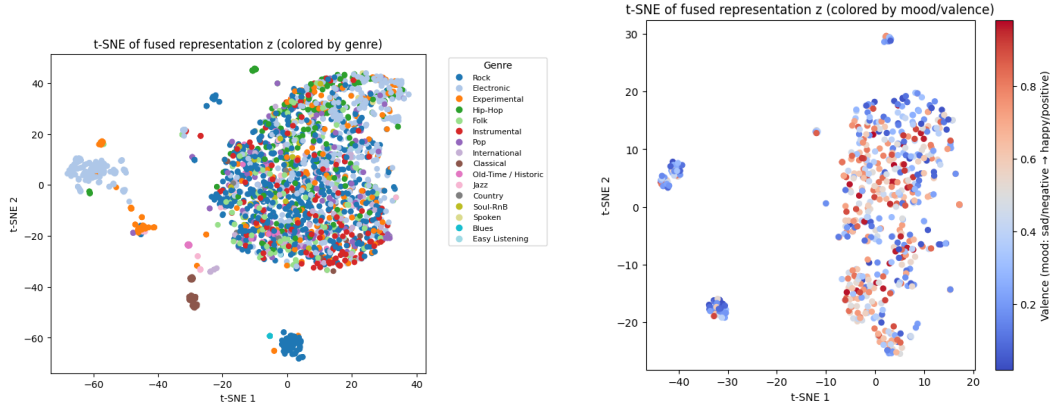


Figure 2: t-SNE of the fused representation z on FMA-medium validation tracks. **Left:** coloured by genre. Several compact single-genre clusters (e.g. Rock, Electronic, Classical) separate from one large mixed cluster. **Right:** the subset of tracks with Echo Nest valence scores, coloured by valence. There is no visible valence gradient, so the representation does not organise tracks by mood.

accuracy indicates that it mostly predicts the most frequent genres. We analyse why the graphs carry so little information in Section 6.

Task 3: cross-attention fusion is the strongest variant. Each unimodal branch is weak on its own: BERT-only reaches 0.171 macro-F1 and GNN-only 0.098, which is consistent with Task 2. Both fusion models beat both unimodal models on every metric. This suggests that the tag text and the audio-structure graph carry at least partly complementary information about genre. Cross-attention improves macro-F1 over early concatenation by 0.063 (0.269 vs. 0.206) and accuracy by 1.2 points. Early concatenation has the higher mean AUC-PR (0.291 vs. 0.276), so the ranking between the two fusion schemes depends on the metric. Given that these are single runs, we regard the gap between the two fusion schemes as suggestive rather than established. The advantage of fusion over either modality alone is larger and more consistent.

5.2 Qualitative analysis

Embedding structure. A t-SNE [van der Maaten and Hinton, 2008] projection of z coloured by genre is depicted in Figure 2 (left). While there are a number of small, close-knit groups (Rock, Electronic, Classical, and less prominent Hip-Hop and Experimental) off to the sides, there is a larger central grouping where genres are generously intermixed. One guess is that the compact clusters are from tracks with informative tag text, and the mixed cluster is from the many tracks with an input of “no tags” and thus relying on the weak graph branch. This hasn’t been verified yet using coloration of the projection by text availability. When colouring tracks using Echo Nest’s valence score (right), there is no obvious mood gradient, as there was no emotion signal used during training.

Attention case studies. In all three samples of text to be validated for review, the text entered is “no tags”. The two tokens were almost equally split (e.g., 0.265 and 0.265), with the three tracks misclassified as follows: 1 Experimental track was classified as Rock, and 2 Rock tracks were classified as Electronic. They had between 3 and 4 nodes and between 4 and 6 edges in their graphs. These examples illustrate that the absence of meaningful text to attend to means that the prediction depends on a four-node graph.

6 Discussion and limitations

The graphs are an information bottleneck. Each segment graph of the FMA has about 4 nodes, each a 12-dimensional mean chroma vector, which is about 48 numbers per track (30 s clips and 7.5 s windows). The CNN, on the other hand, displays a full 128×1300 log-mel spectrogram. Chroma also does not adjust for timbre and rhythm, which is important for genre. If a segment has a mean-pooling

value of over 7.5 s, the majority of the harmonic movement inside the segment is removed, and similarity edges are unlikely to bring much beyond the temporal chain if there are only four nodes. Lastly, GraphSAGE doesn't consider the cosine edge weights. Task 2 gap is thus considered not as a criticism of structural representations of music as such, but rather of the particular type of graph construction used here. Next steps are clear and include shorter or beat-synchronous segments (code has this capability), richer node features that include mel, chroma and MFCC statistics, and an edge-aware encoder.

Task definitions. Task 1 uses the input provided in its own labels and does not attempt to assess an understanding of text. Furthermore, the random mask for validation clips is sampled from a common mask generator, and thus the validation inputs will vary with each epoch, thus adding noise to the Task 1 curve. Task 3 is intended to be multiple labels (valence/arousal regression) but is actually run as a single label (genre classification), and the emotion head is not applicable. In FMA, the free-text tags are often empty, and therefore the text branch lacks a lot of information. The impact on the number of tracks affected and the fact that sometimes the tags contain the genre name itself (partly leaking the label) is not measured.

Statistical reliability. All results are obtained from a single seed, measured at the end of the last epoch on the validation split, and are not error bars. The CNN's swing between 0.36 and 0.28 macro-F1 for the past few epochs indicates that the epoch-to-epoch difference is similar to some of the differences reported. The test splits were not evaluated. Because hyperparameters were chosen on the same validation split used for reporting, the reported scores are likely somewhat optimistic. Some of the validation items may be duplicated when undecodable tracks are replaced with neighbouring tracks.

Broader impact. Improvements in the accuracy of music tagging and genre recognition can assist music cataloging, search, and discovery, even for independent artists like FMA. They can also contribute to popularity bias in recommendations, and a narrow, Western view of music genres can misrepresent music outside of the genre. Our models are small, trained on openly licensed data, not deployed – see no direct path to misuse.

7 Conclusion

We built a hybrid system that encodes audio-structure graphs with GraphSAGE and tag text with BERT, and combines them through a graph-to-text cross-attention layer. On FMA-medium genre classification, fusion outperforms either modality alone, and cross-attention gives the best macro-F1. The structure-only branch, however, falls well short of a simple spectrogram CNN, because the current graphs contain only about four low-dimensional nodes per track. The most valuable next steps are (1) richer and finer-grained segment graphs with edge-aware GNNs, (2) evaluation on held-out test splits over several seeds, (3) multi-label tagging and emotion targets for the fusion model, as originally planned, and (4) the contrastive audio-graph-caption retrieval task on MusicCaps.

References

- Andrea Agostinelli, Timo I. Denk, Zalán Borsos, Jesse Engel, Mauro Verzetti, Antoine Caillon, Qingqing Huang, Aren Jansen, Adam Roberts, Marco Tagliasacchi, et al. MusicLM: Generating music from text. *arXiv preprint arXiv:2301.11325*, 2023.
- Keunwoo Choi, György Fazekas, and Mark Sandler. Automatic tagging using deep convolutional neural networks. In *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)*, 2016.
- Michaël Defferrard, Kirell Benzi, Pierre Vandergheynst, and Xavier Bresson. FMA: A dataset for music analysis. In *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)*, 2017.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pages 4171–4186, 2019.

- Benjamin Elizalde, Soham Deshmukh, Mahmoud Al Ismail, and Huaming Wang. CLAP: Learning audio concepts from natural language supervision. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.
- Matthias Fey and Jan Eric Lenssen. Fast graph representation learning with PyTorch Geometric. In *ICLR Workshop on Representation Learning on Graphs and Manifolds*, 2019.
- Jonathan Foote. Automatic audio segmentation using a measure of audio novelty. In *Proceedings of the IEEE International Conference on Multimedia and Expo (ICME)*, 2000.
- William L. Hamilton, Rex Ying, and Jure Leskovec. Inductive representation learning on large graphs. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Qingqing Huang, Aren Jansen, Joonseok Lee, Ravi Ganti, Judith Yue Li, and Daniel P. W. Ellis. Mu-Lan: A joint embedding of music audio and natural language. In *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)*, 2022.
- Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations*, 2017.
- Edith Law, Kris West, Michael Mandel, Mert Bay, and J. Stephen Downie. Evaluation of algorithms using games: The case of music tagging. In *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)*, pages 387–392, 2009.
- Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.
- Jiasen Lu, Dhruv Batra, Devi Parikh, and Stefan Lee. ViLBERT: Pretraining task-agnostic visual-linguistic representations for vision-and-language tasks. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- Brian McFee and Daniel P. W. Ellis. Analyzing song structure with spectral clustering. In *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)*, 2014.
- Brian McFee, Colin Raffel, Dawen Liang, Daniel P. W. Ellis, Matt McVicar, Eric Battenberg, and Oriol Nieto. librosa: Audio and music signal analysis in Python. In *Proceedings of the 14th Python in Science Conference (SciPy)*, 2015.
- Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, et al. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- Jordi Pons, Oriol Nieto, Matthew Prockup, Erik M. Schmidt, Andreas F. Ehmann, and Xavier Serra. End-to-end learning for music audio tagging at scale. In *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)*, 2018.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2021.
- Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.
- Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9:2579–2605, 2008.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. Graph attention networks. In *International Conference on Learning Representations*, 2018.

Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, et al. Transformers: State-of-the-art natural language processing. In *Proceedings of EMNLP: System Demonstrations*, pages 38–45, 2020.

Minz Won, Andres Ferraro, Dmitry Bogdanov, and Xavier Serra. Evaluation of CNN-based automatic music tagging models. In *Proceedings of the Sound and Music Computing Conference (SMC)*, 2020.

A Hyperparameters and reproducibility

Table 2: Hyperparameters used for all reported runs (from `config.yaml`). The same settings are shared across tasks unless stated otherwise.

Setting	Value
Sample rate / hop length	22,050 Hz / 512
Graph node features	12-bin chroma, z-normalised per track, mean over 7.5 s windows
Similarity threshold τ	0.8 (cosine)
GNN	GraphSAGE, 3 layers, hidden 128, dropout 0.2, mean pooling
Text encoder	bert-base-uncased, fully fine-tuned, max length 256
Fusion attention dim d_a	128
CNN baseline input	128-band log-mel, 1,300 frames
Optimiser	AdamW, weight decay 0.01
Learning rate	2×10^{-5} (BERT; all of Task 1), 10^{-3} (new modules)
Batch size / epochs / seed	32 / 20 / 42
Task 1 tags / mask ratio	top 50 / 0.5
Genre classes	16 (FMA-medium top-level)

Commands. After downloading the datasets and setting paths in `config.yaml`, each task is run with `python src/train.py --task N` for $N \in \{1, 2, 3\}$. Task 2 also trains the CNN baseline, and Task 3 trains all four fusion variants and produces the t-SNE and case-study outputs. The mood t-SNE is produced by `python src/mood_tsne.py` after Task 3 has finished, since it loads the Task 3 checkpoint. `--max-samples` and `--epochs` allow quick smoke tests.

Compute. Tasks 1 and 2 were run on a single NVIDIA GeForce RTX 3050 GPU (4 GB VRAM) with 8 GB of system RAM. Task 1 took approximately 7 hours, and Task 2 (the GNN and the CNN baseline, including audio decoding and caching of graphs and spectrograms) approximately 12 hours. Task 3 (all four fusion variants) was run on a single NVIDIA GeForce RTX 4070 Ti Super GPU (16 GB VRAM) with 64 GB of DDR5 RAM and took approximately 40 minutes. All times are approximate wall-clock figures, for a total of roughly 20 GPU-hours.

B Training curves

Figure 3 shows validation macro- and micro-F1 per epoch for each task.

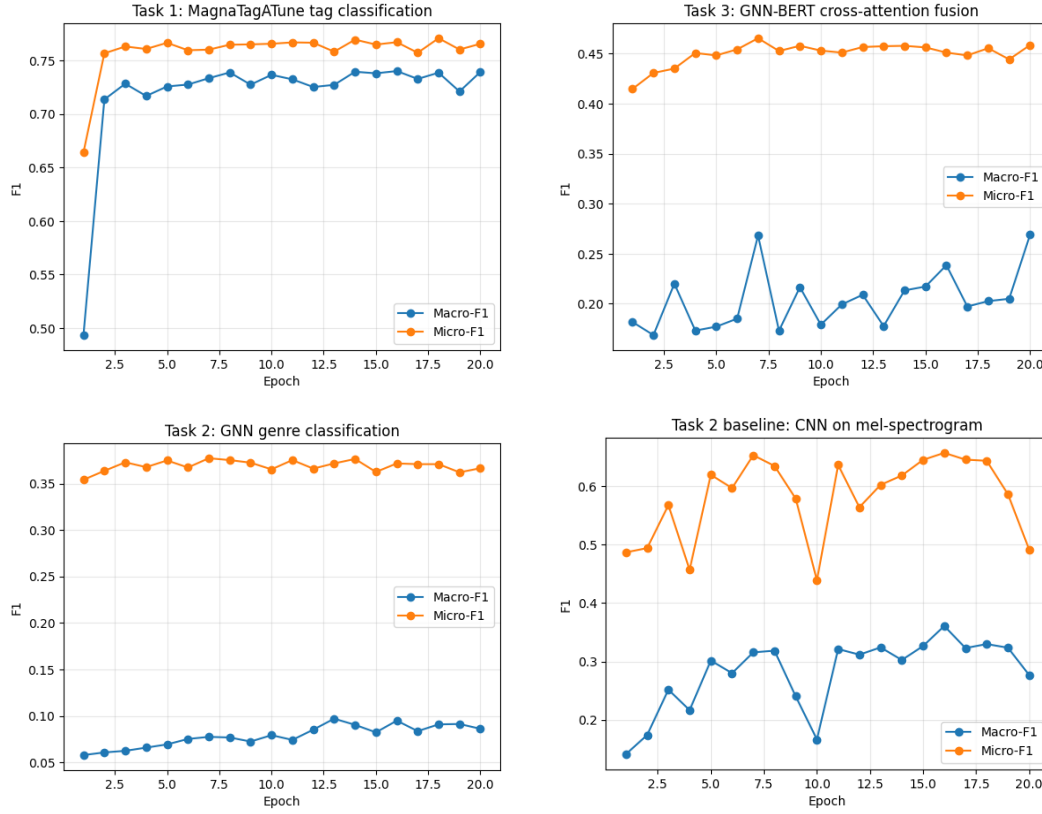


Figure 3: Validation macro- and micro-F1 per epoch. **Top:** Task 1 BERT tag completion (left) and Task 3 cross-attention fusion (right). **Bottom:** Task 2 GNN (left) and CNN baseline (right). The CNN varies strongly from epoch to epoch, so final-epoch numbers understate its best validation performance.

C Planned Task 4: contrastive audio-graph–caption retrieval

Task 4 targets MusicCaps [Agostinelli et al., 2023], the only dataset in our set with natural-language captions. A dual encoder maps a segment graph through the GNN and a caption through BERT, and linear projections map both into a shared 128-dimensional space, giving $\mathbf{u}_i$ and $\mathbf{v}_i$ after ℓ_2 normalisation. Training uses a symmetric InfoNCE loss [van den Oord et al., 2018, Radford et al., 2021] with temperature $t = 0.07$:

$$\mathcal{L} = -\frac{1}{2B} \sum_{i=1}^B \left[\log \frac{\exp(\mathbf{u}_i^\top \mathbf{v}_i/t)}{\sum_j \exp(\mathbf{u}_i^\top \mathbf{v}_j/t)} + \log \frac{\exp(\mathbf{u}_i^\top \mathbf{v}_i/t)}{\sum_j \exp(\mathbf{u}_j^\top \mathbf{v}_i/t)} \right]. \quad (2)$$

Evaluation would use Recall@ K in both retrieval directions. Task 4 is not implemented in the released code; only its planned hyperparameters remain in `config.yaml`. We therefore make no claims about it.

D Licenses of existing assets

MagnaTagATune [Law et al., 2009]: CC BY-NC-SA 3.0. FMA [Defferrard et al., 2017]: meta-data CC BY 4.0; audio under the Creative Commons licenses chosen by each artist; code MIT. MusicCaps [Agostinelli et al., 2023]: CC BY-SA 4.0 (not used in reported experiments). bert-base-uncased [Devlin et al., 2019]: Apache 2.0. Hugging Face Transformers: Apache 2.0. PyTorch: BSD-3-Clause. PyTorch Geometric: MIT. librosa: ISC.

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Justification: The abstract and Section 1 state the results as measured, including the negative Task 2 result, and note that all numbers come from single runs on validation splits and that Task 4 was not executed.

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2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification: Section 6 discusses the graph information bottleneck, the self-derived Task 1 inputs, the deviation of Task 3 from its planned scope, sparse FMA text, single-seed validation-only evaluation, and track substitution.

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